CHAPTER 9

Gallstone: a factor for acute pancreatitis

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Introduction

Pancreatitis is one of the major inflammatory diseases associated with the exocrine gland, broadly classified as acute and chronic pancreatitis. Clinical symptoms of acute pancreatitis (de Roodt et al., 2009) are the sudden onset of stomach discomfort (upper abdominal pain) and an increased level of pancreatic enzymes in the blood or urine. Attacks are generally innocuous, although serious assaults can result in shock and renal failure. Morphologically, acute pancreatitis is characterized by lesions that, in moderate conditions, comprise scattered patches of fat necrosis and pancreatic edema, while in severe conditions, it can encompass significant peripancreatic and intrapancreatic fat necrosis, parenchymal necrosis, and excess bleeding, followed by death due to respiratory insufficiency (Bollen, 2016). Clinically, chronic pancreatitis is characterized by recurrent or persistent stomach pain, though it can also happen without discomfort in some circumstances. Sclerosis with the destruction and irreversible loss of exocrine components is among the morphologic alterations associated with chronic pancreatitis (Beyer et al., 2020; Kleeff et al., 2017). More than half of the cases of acute pancreatitis are due to cholelithiasis, which predominantly blocks the bile duct and sets off a series of inflammatory responses (Lankisch et al., 2015; Wang et al., 2009). Acute pancreatitis can lead to hospitalization or death, so it becomes pertinent to identify and characterize the disease at the earliest possible time. The Glasgow scale, Ranson's criteria, and Acute Physiology and Chronic Health Evaluation (APACHE II) systems help achieve the early detection of the disease. APACHE systems are most frequently used in grading the severity level because of their higher sensitivity and quicker analysis (Aerts & Penninckx, 2003; Chatzicostas et al., 2002).

Gallstone Formation, Diagnosis, Treatment, and Prevention
DOI: https://doi.org/10.1016/B978-0-443-16098-1.00015-1
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Epidemiology

Approximately 20% of adult Europeans are suffering from gallstones (Carey & Paigen, 2002). Gallstones-related pancreatitis is more common in Central and South American Hispanic groups and among American-Hispanics of Native American ancestry (Miquel et al., 1998). Among the Asian and African populations, the prevalence is comparatively less (5%–15%) (Stokes et al., 2011; van Erpecum, 2006; Völzke et al., 2005). The prevalence of gallstone-related acute pancreatitis increases with age and is more common in women than in men (Tsai et al., 2006). Various epidemiological studies have investigated the factors contributing to the cause of gallstone formation, leading to acute pancreatitis (Chait et al., 1979). Obesity, lack of physical activity, consumption of fatty food, and type 2 diabetes mellitus are some of the predisposing and exogenous factors that contribute to the formation of gallstones, followed by pancreatitis (Buch et al., 2007, 2010; Lammert & Sauerbruch, 2005; Nervi et al., 2006; Weikert et al., 2010). Various genetic factors frequently involved in gallstone formation are the gene coding the hepatobiliary cholesterol transporter (ABCG8p.D19H), (Adams, 2007; Buch et al., 2007) mutations in genes like ABCG8, UGT1A1, and ABCB4, which contribute to the metabolism of gallstones and can also be factored into pancreatitis (Buch et al., 2010; Dimarino, 2000).

Etiology

Various theories proposed to explain the causes that result in gallstone-related acute pancreatitis are as follows: (1) common channel hypothesis, (2) pancreatic reflux hypothesis, and (3) ductal hypertension hypothesis (Acosta & Ledesma, 1974).

Common channel hypothesis

According to this hypothesis, the reflux of bile into the pancreatic duct, which is caused by blockage of the common pathway or channel for the hepatic and pancreatic ducts, is the main cause of acute pancreatitis. Various experiments on dogs showed that forcible bile injection into the pancreatic ducts causes pancreatic inflammation (Kahl & Mayer, 2012). Although the "common channel" concept is one of the most widely accepted hypotheses in the field and appears to make sense from a mechanical perspective, a lot of experimental and clinical evidence contradicts its presumptions. According to anatomical research, the pancreatic

duct and common bile duct do not communicate well enough to allow biliary reflux into the pancreatic duct; an impacted gallstone would likely block both the common bile duct and the pancreatic duct. Pancreatic secretory pressure would still be higher than biliary pressure even if there was anatomical communication, causing pancreatic juice to flow into the bile duct rather than bile into the pancreatic duct (Jones et al., 1987).

Duodenal reflux hypothesis

The reflux of duodenal material into the pancreatic duct is a second possible mechanism of pancreatitis brought on by gallstone transit. This hypothesis has been utilized in dogs to induce acute pancreatitis; here, a closed loop of the duodenum is created with subsequent induction of pancreatitis due to the reflux of contents (Byrne et al., 1964). In humans, there is a rare parallel where occlusion of the afferent loop following gastrectomy or gastroenterostomy may, on occasion, result in acute pancreatitis (Lee & Sarosi, 2011). However, various defense mechanisms surround the normal pancreatic duct to stop this reflux (Moreira et al., 1992).

Ductal hypertension hypothesis

Lerch et al. examined how obstruction at various locations in the pancreaticobiliary ductal tree affected opossums' development of pancreatitis. They demonstrated that, in the animal models, blockage of the main pancreatic duct alone is sufficient to elicit pancreatitis and that the severity of the disease was unaffected by separate closure of the common bile duct or the common biliopancreatic channel. Other research has demonstrated that maintaining secretion in the context of a blocked pancreatic duct causes more damage. Several additional, less frequent, causes of acute pancreatitis, including ampullary tumors, helminthic infestations, and endoscopic retrograde cholangiopancreatography (ERCP), would also result in ductal hypertension. Indeed, acute pancreatitis can be brought on by injecting a variety of substances into the pancreatic duct at pressures above physiological (Vítek & Carey, 2012).

Pathophysiology

The pathophysiology of gallstone-related pancreatitis is complex; it starts with gallstone formation, has impacts on the acinar cells, and finally, has the immune responses that exacerbate the whole process (Biddinger et al., 2008).

Gallstone formation

Bile plays an important role in the formation of gallstones. Over 90% of the liquid in bile is water; bile juice is dark green to yellowish-brown in color. Three primary lipids found in bile are cholesterol, phospholipids, and bile salts. In addition to bile lipids, bile juice also includes trace amounts of proteins and inorganic salts. Gallstones are usually separated into two categories: cholesterol and pigment gallstones, with two distinct etiologies based on chemical makeup and macroscopic appearance (Buch et al., 2007; Lyons & Wittenburg, 2006).

Cholesterol gallstone

Failure of biliary cholesterol homeostasis results in the production of cholesterol gallstones. Various factors responsible for cholesterol gallstone formation are hepatic cholesterol hypersecretion, intestinal and gallbladder hypermotility, supersaturated bile, and rapid phase transition.

Pigment gallstone

Pigment stone formation occurs due to improper bilirubin metabolism. Patients with black or brown pigment stones have elevated levels of unconjugated bilirubin in their bile. They are mainly of two types: black pigment stones and brown pigment stones. Black pigment stones are formed in the gallbladder and are generally not infected by microorganisms. They are formed during events that increase the amount of bilirubin, like ileal diseases and hemolytic anemias. Black pigment stones are made of calcium bilirubinate, either in its pure form or as part of polymer-like complexes with copper, calcium, and unconjugated bilirubin. Brown pigment stones mostly contain calcium salts of unconjugated bilirubin and may also contain trace amounts of other components like cholesterol, fatty acids, pigment, mucin glycoproteins, bile salts, phospholipids, and bacterial remnants. All areas of the biliary tree, particularly bile ducts, are susceptible to brown pigment stone formation. Both biliary infection (particularly with *Escherichia coli*) and bile stasis caused by bile duct obstruction are crucial circumstances. To convert bilirubin glucuronide into unconjugated bilirubin, *E. coli* produces enzymes like glucuronidase, phospholipase A1, and conjugated bile acid hydrolase. Water-insoluble unconjugated bilirubin reacts with calcium to form calcium bilirubinate at its carboxyl radical, which leads to the formation of brown pigment stones (Wasmuth et al., 2006).

Cellular level changes

The pancreatic acinar cells are a major component of the exocrine pancreas. Digestive enzymes are produced, stored, and secreted by them. Enzymes are secreted into the pancreatic lobules and then into the duodenum. Getting signals like cholecystokinin (CKK) and secretin, which are released in response to food, increases calcium signaling, ultimately leading to secretion (Hegyi & Petersen, 2013; Sluga et al., 2021). Acute pancreatitis is a clinical condition that generally begins when these enzymes are prematurely activated within pancreatic acinar cells and subsequently leads to a cascade of intracellular signaling, ultimately leading to acinar cell death and edema (Leung & Ip, 2006). Pancreatitis caused by gallstones occurs due to pancreatic juice outflow issues brought on by the entrapment of the common bile duct by stones into the papilla, leading to pancreatic inflammation and damage. Acinar cell necrosis and damage are the main reasons that drive the process of inflammatory damage. Various hypotheses are being put forward, like intracellular calcium, premature zymogen activation, failure of autophagy, and the immune responses from the damaged acinar cells, which can explain the possible reasons for the pancreatic acinar cell necrosis and inflammation that result from it (Kimura et al., 2010).

Excess calcium

A major feature of acute pancreatitis is the pathological elevation of Ca^{2+} concentration in acinar cells, which causes pro-cell death and pro-inflammatory pathways including premature trypsinogen activation, activation of nuclear factor-B (NF-B), and mitochondrial dysfunction (Jakkampudi et al., 2016; Krüger et al., 2000; Maléth & Hegyi, 2016). Ca^{2+} ions are released from the ER during a physiological state as a part of a signaling pathway that starts zymogen exocytosis and encourages the generation of ATP in the mitochondria. The smooth endoplasmic reticulum Ca^{2+} channels (SERCAs) transport Ca^{2+} ions back into the ER, and the plasma membrane Ca^{2+} channels (PMCAs) exude Ca^{2+} out of the cell, but the increase in cytosolic Ca^{2+} concentration is only momentary, as two ATP-dependent calcium channels quickly remove the cytosolic calcium (Criddle et al., 2007). Through the inositol 1,4,5-trisphosphate receptor signaling pathway, bile acids (which reflux due to duct obstruction) upset this equilibrium and result in a widespread, persistent pathogenic increase of cytosolic Ca^{2+}, which in turn causes further release of calcium from intracellular stores (Gerasimenko et al., 2013; Wen et al.,

2015). There is also the simultaneous entry of calcium from extracellular space into the cytosol through ORAI1 channel proteins, which further adds to the toxicity (Feng et al., 2019; Lur et al., 2011). Gallstone pancreatitis causes ductal blockage, which is considered to promote Ca^{2+} entrance from the extracellular space through PIEZO1, a pressure-activated plasma membrane mechanoreceptor with cation channel characteristics (Romac et al., 2018). The loss of the membrane potential required to produce ATP is brought on by the mitochondrial permeability and transition pores opening in a high-conductance state as a result of the excess cellular Ca^{2+} concentration (Elmunzer et al., 2016; Mukherjee et al., 2016). The lack of ATP production leads to the failure of ATP-dependent SERCAs and PMCAs to clear the excess calcium inside the cytosol, hence interfering with cytoprotective systems like autophagy and the UPR. Additionally, premature activation of zymogens by various mechanisms (which will be discussed later) leads to necrosis of acinar cells (Maléth & Hegyi, 2014; Pallagi et al., 2020).

Premature trypsinogen activation

Another significant pathogenic cellular process is premature trypsinogen activation. Acinar cell necrosis may result from a common occurrence. A variety of pancreatic traumas, including alcohol, trauma, and obstruction of the pancreatic duct, can cause the lysosome to fuse with the zymogens inside the acinar cells, known as colocalization (Dawra et al., 2011). Once the lysosomes fuse with the zymogen vesicles containing the inactive trypsinogen, the lysosomal enzyme cathepsin B converts the inactive trypsinogen to trypsin (Halangk et al., 2000). The released trypsin causes autodigestion of the inner cellular matter and the release of cathepsin B, which causes necrosis *via* receptor-interacting protein kinase (RIP), mainly RIP1-RIP3, and mixed lineage kinase (MLK). The MLK proteins are converted to their phosphorylated form by RIP1-RIP3 l, and the phosphorylated form leads to necrosis of the cell (Han et al., 2011; Kuhl et al., 2007).

Autophagy failure

Impaired cytoprotective mechanisms due to excess bile and intracellular calcium also have a role in the pathogenesis of acute pancreatitis. A cytoprotective mechanism called macroautophagy processes and recycles numerous worn-out cytoplasmic contents that are flawed or damaged (Antonucci et al., 2015). Due to trypsinogen activation, ER stress resulting from the

failure of the endoplasmic reticulum to process an excessive amount of protein and mitochondrial dysfunction brought on by defective autophagy-stressed endoplasmic reticulum triggers transcription of CHOP proteins, which cause cell death and the release of inflammatory mediators (Han et al., 2011; Lugea et al., 2017; Wu et al., 2016; Zelic et al., 2018).

CCK and bile acid trigger the release of calcium from the endoplasmic reticulum through the inositol 1,4,5-trisphosphate receptor (Ins(1,4,5)P3R) in the acinar cell endoplasmic reticulum. Calcium enters the cell from the extracellular environment through the calcium release-activated calcium channel protein 1 (ORAI1), which opens as a result of the low calcium concentration in the endoplasmic reticulum (ER). The elevated calcium concentration that occurs is pathological. Loss of membrane potential occurs across the mitochondrial membrane as a result of the opening of mitochondrial permeability transition pores to a high-conductance state in response to calcium increase, which leads to cell necrosis. The excessive calcium, along with bile salts and CCK, leads to premature zymogen activation, which activates the RIP3–RIP1 complex, leading to the activation of phosphorylated mixed lineage kinase domain-like (MLKL) and necrosis. Bile and excessive calcium levels also prevent autophagy, which causes the accumulation of misfolded proteins leading to endoplasmic reticulum stress and the expression of ATF6, which further promotes transcription of CHOP proteins, ultimately leading to cell death (Fig. 9.1).

Immune response

Chemokines, cytokines, and other adhesion molecules released by injured acinar cells help draw immune cells to the damage site and facilitate their infiltration (Fig. 9.2). When immune cells enter the pancreas, the contents of necrotic and damaged cells are discharged, activating monocytes and neutrophils to spread the inflammation. The activated neutrophils, by various enzymatic actions, cause intra-acinar cell trypsinogen activation and an increase in oxidative stress. Additionally, neutrophils create neutrophil extracellular traps, which are adhesive webs made of granular proteins, DNA, and histones and can clog ducts and trigger pro-inflammatory signals. In acute pancreatitis, activated monocytes play a key role in the development of tissue damage and systemic inflammation. Damage-associated molecular patterns (DAMPs), biological components released from necrotic acinar cells, are significant mediators of monocyte activation. DAMPs attach to several immune cell receptors, which then mediate their effects. For instance, the DAMPs heat shock protein 70 and high

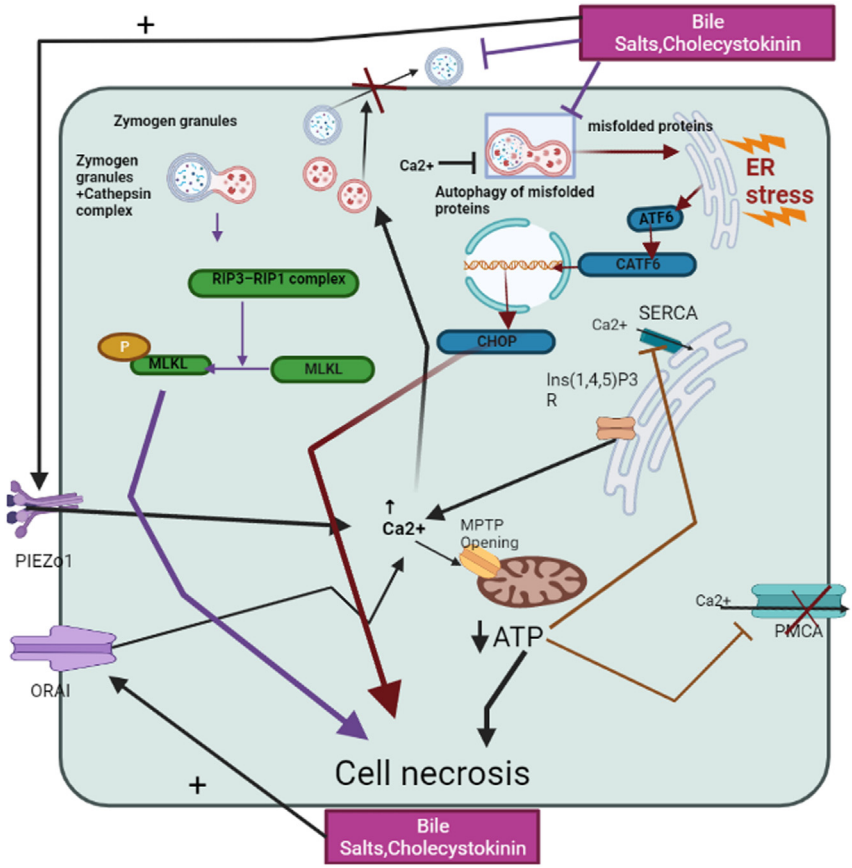

Figure 9.1 Pathway showing the cell death triggered by the Cholecystokinin (CCK), and bile acid.

mobility group box protein signal *via* toll-like receptors and the expression of pro-inflammatory cytokines, chemokines, and adhesion molecules *via* the NF-κB pathway (Zhou et al., 2010). Other DAMPs that bind to P2X7 receptors, such as ATP and NAD, cause the inflammasome to become active. Pro-IL-1β and pro-IL-18 are then proteolytically cleaved into their physiologically active counterparts. TNF, IL-1β, IL-6, and IL-18 are just a few of the pro-inflammatory cytokines that are amplified through these pathways. Due to the inflammatory action of the immune cells, macrophages at distant organs are also activated, which worsens systemic inflammation and distant organ harm (Bhatia et al., 2000; Malla et al., 2016).

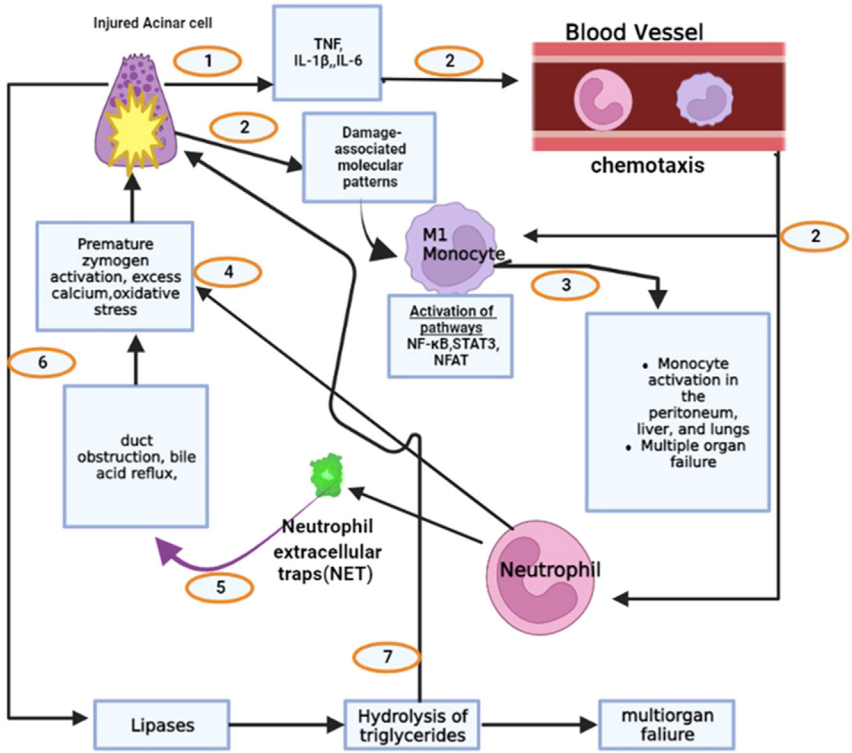

Figure 9.2 Immune cell infiltration at the damage site triggered by injured acinar cells.

Disease models

Over the past few decades, various disease models have improved the understanding and elucidation of the pathophysiology and intricate mechanisms involved in disease development in many species, including pigs, dogs, cats, rabbits, and zebrafish. Zebrafish as well as opossums have been utilized as pancreatitis models; however, the majority of researchers choose rodents because they are more affordable and easy to maintain. Genetic modification is also possible in mice and rats to develop transgenic or knockout animals, and reagents are also readily accessible (Lerch et al., 1992). Over the years, various murine models have been developed, and some of the well-known models are calcineurin, bile salt duct infusion, diet-induced models, and basic amino acid-induced models (Gorelick & Lerch, 2017; Yang et al., 2020). Gallstone-related pancreatitis accounts for 50% of the total cases (Yang et al., 2020). Gallstones can either allow bile to reflux into the pancreatic duct or temporarily impede

pancreatic output. These two theories were supported by autopsy results in various patients and meticulously executed animal experiments. Pancreatic duct blockage caused by the impacted gallstones would stop pancreatic secretion and cause pancreatitis through inflammatory responses (Lerch et al., 1993; Vigna et al., 2011), which causes damage due to free radicals generated in the acinar cells. The main models that have been used for cholelithiasis-related acute pancreatitis are the bile duct ligation model and the closed duodenal loop acute pancreatitis model (Liu et al., 2022; Pedro-Botet et al., 2020).

Overview of models in acute pancreatitis

This is a murine model in which acute pancreatitis is induced by ligating the distal bile duct at the level of the duodenum. This model mimics the situation where gallstones cause blockage in the ampulla of Vater and hence lead to reflux of bile throughout the pancreatic duct (Yang et al., 2020). The model induces inflammation, edema, and hyperamylasemia, which are features of pancreatitis (Ohshio et al., 1991). Additionally, this model causes rat liver cholangitis and obstructive hepatic cholestasis. However, some modifications were made to this model to improve it, including the addition of pancreatic duct ligation and secretory stimulation with caerulein or sodium taurocholate. According to studies, apoptosis is the main mechanism of cell death in the rat duct ligation model. It is documented that blockage of bile flow into the intestine results in small bowel bacterial overgrowth; this model might be appropriate for the investigation of bacterial translocation (Nieuwenhuijs et al., 2000).

Duct ligation model

This is a murine model, in this acute pancreatitis is induced by ligating the distal bile duct at the level of the duodenum, this model mimics the situation where gallstones cause blockage in the ampulla of Vater and hence lead to reflux of bile throughout the pancreatic duct (Yang et al., 2020). The model induces inflammation, edema, and hyperamylasemia which are features of pancreatitis (Ohshio et al., 1991). Additionally, this model causes rat liver cholangitis and obstructive hepatic cholestasis. However, some modifications were made to this model to improve it, including the addition of pancreatic duct ligation and secretory stimulation with caerulein or sodium taurocholate. According to studies, apoptosis is the main mechanism of cell death in the rat duct ligation model. It is

documented that blockage of bile flow into the intestine results in small bowel bacterial overgrowth, this model might be appropriate for the investigation of bacterial translocation (Nieuwenhuijs et al., 2000).

Closed duodenal loop acute pancreatitis model

This is a rat-based murine model wherein duodenal pancreatic reflux is avoided in a physiologically normal environment by confirming a pressure gradient between the common bile duct and the duodenum. By altering this typical state, the closed duodenal loop model raises intraduodenal luminal pressure, resulting in reflux and acute pancreatitis. This type is distinguished by the duodenum's ligation both proximally and distally to the common biliopancreatic duct junction. Interlobular edema and gradually hemorrhagic pancreatic necrosis are the histological alterations confirmed by this animal (Xie et al., 2019). Some scientists proposed that the cannulation of bile is typically channeled into the jejunum from the bile duct to keep it out of the duodenum and prevent it from contributing to the pancreatitis that results from this. Some scientists pressurize the duodenum and inject sterilized or infected bile, resulting in severe AP. Some researchers changed the original model to a temporary duodenal ligation because this model is linked to a high rate of rat death. This model is also suitable to study bacterial translocation during acute pancreatitis (Ceranowicz et al., 2015).

The common biliopancreatic duct is located and bound at the level of the duodenum using aseptic procedures, which involve exposing the duodenum during a laparotomy. Fig. 9.3 illustrates where the duodenum's closed loop is located. Aseptic approaches involve a laparotomy to expose the duodenum, identification of the common biliopancreatic duct, and obstruction of the duodenum with the application of two ligatures, one proximally to the duct and the other distally.

Diagnosis

Physical examination

Most of the patients with gallstone acute pancreatitis show typical symptoms of acute pancreatitis, like sudden epigastric pain or right upper quadrant abdominal pain, which is unrelenting and radiates to the back, and nausea and vomiting, which accompany the other symptoms (Attasaranya et al., 2008; Lankisch et al., 2015).

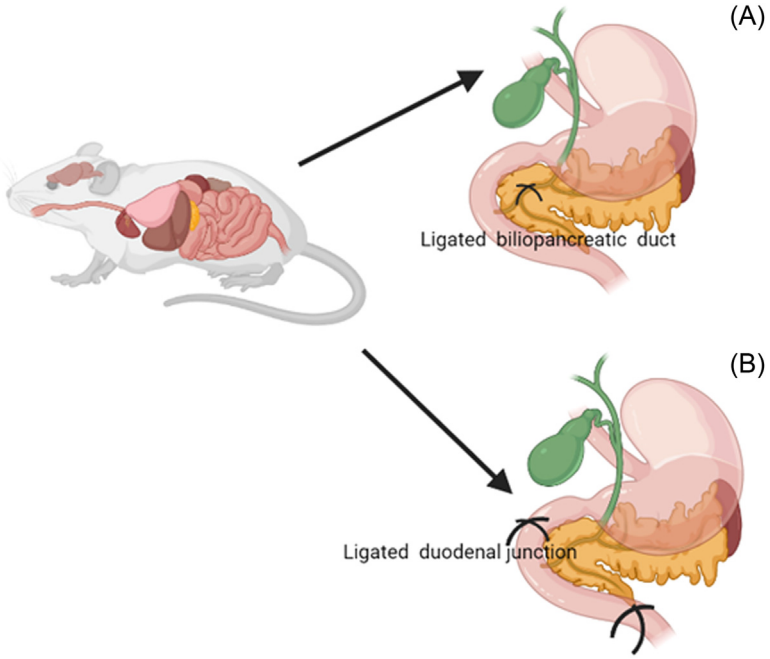

Figure 9.3 Image displaying the ligation site of rat's common biliopancreatic duct (A) and site of duodenum's closed loop (B).

Biochemical markers

Patients with acute pancreatitis very often show an elevated level of amylase and alanine transferase enzymes, which can be very often related to the nausea and vomiting the patient experience, and they are found at higher levels in patients suffering from gallstone-related pancreatitis (Abu-El-Haija et al., 2018; Fogel & Sherman, 2003; Ogrinc et al., 1999), but tests do not indicate the intensity of the disease (Yadav et al., 2007). A complete blood count has shown a significant increase in leukocytes and hematocrit values to some extent, which helps to determine the severity of the disease (Auti et al., 2022). Liver function tests have also shown increased bilirubin and transaminase levels, since the underlying disease in GSP may involve a biliary blockage, albeit temporary in most circumstances. If cholestasis persists, alkaline phosphatase and gamma-glutamyl transpeptidase may be increased (Dholakia et al., 2004; Greenberg et al., 2016; Nathwani et al., 2005).

Imaging techniques

Imaging techniques are a very handy and precise method for the identification of gallstone-related pancreatitis. They identify the main cause of the disease, the associated complications, and the severity of the disease.

Ultrasonography

It is a very convenient and widely used technique to determine the presence of cholelithiasis or sludge. This technique is a confirmation of gallstone-related acute pancreatitis when other biochemical factors are indicative of the disease. The test is highly sensitive but cannot detect stones that are having sizes less than 4 mm (Tenner et al., 1994).

Computed tomography

This is a very precise method that has around 80% accuracy when it comes to the detection of the disease. This technique not only identifies gallstones; however, at the same time, it can also show necrosis and other inflammatory damages. To enhance the quality of the image, contrast agents can be added to improve visualization, but oral contrasting agents must be avoided (Sugiyama & Atomi, 1997).

Magnetic resonance cholangiography

Magnetic resonance cholangiography (MRC) is a recently developed imaging technique that is noninvasive. The technique gives a great resolution of the bile duct and intraductal stones. This imaging technique can produce projectional images that resemble direct cholangiography in terms of detail and look. MRC does not employ ionizing radiation or intravenous contrast, and it is also less operator-dependent. MRC is a helpful initial imaging test for patients with suspected pancreatitis since it is also capable of detecting other biliary pathologies, such as gallbladder stones and pancreatic inflammation (Balthazar et al., 1985).

Management of the disease

The most pivotal step in the management of the disease is to determine the intensity of the disease to provide the ideal treatment. Gallstone-related pancreatitis comes in a variety of intensities, starting with mild pancreatic inflammation, which normally resolves within 24 hours, and there are severe forms that are characterized by severe pancreatic necrosis.

Once the intensity of the disease is high, its management is achieved through a combination of interventions, which include lifestyle changes and therapeutic and surgical removal as per the requirements.

Indices of intensity

Various indices are used to grade the intensity of gallstone pancreatitis. The oldest and one of the most common indexes is the Ranson index, which is calculated based on 11 parameters listed in Table 9.1. A Ranson score of 3 or more is associated with serious acute pancreatitis (Chen et al., 2015).

APACHE II is another widely used index and has been extensively used on people with acute pancreatitis. According to reports, APACHE-II at 48 hours has a better positive predictive value than the Ranson score, and it can be calculated at any point during admission. An APACHE score of 8 or more is indicative of severe pancreatitis. The various parameters evaluated are given in Table 9.2 (Ong & Shelat, 2021).

To grade the severity of acute pancreatitis based on morphologic characteristics such as gland enlargement, peripancreatic inflammation, fluid collections, retroperitoneal gas, and degree of pancreatic necrosis, Balthazar and colleagues developed the CT severity index. This index can only be used for patients for whom the disease is already established (Knaus et al., 1981).

Table 9.1 List of parameters evaluated for the APACHE II system.

S. no.	
1	Age
2	History of immunocompromise
3	Rectal temperature
4	Mean arterial temperature
5	Heart rate
6	Respiratory rate
7	Oxygenation
8	Serum sodium
9	Serum potassium
10	Serum creatinine
11	Hematocrit
12	Arterial pH
13	White blood count

Table 9.2 Various parameters for various pancreatitis in the Ranson system.

Parameters	Other pancreatitis	Gallstone pancreatitis
On admission		
White blood cell count	>16,000	>18,000
Serum glucose (mg/dL)	>220	>220
Serum AST (IU/L)	>250	>250
Serum LDH (IU/L)	>350	>400
Within 48 h		
Base deficit (mmol/L)	>4	>5
Hematocrit decrease	>10%	>10%
BUN increase (mmol/L)	>5	>2
PaO2 (mm Hg)	<60	<60
Sequestration of fluids (L) >4	>6	>4
Serum calcium (mg/dL)	<8	<8

Nutritional support

During acute pancreatitis of any form, there is a rapid metabolism of fats and proteins due to the hypercatabolic state. Nutritional support targets providing adequate calories for the requirements inside our body and, at the same time, reducing the inflammatory burden. Evidence supports the introduction of a substantial solid low-fat diet in individuals with mild or moderately severe pancreatitis, but this must be done with caution in patients having pain relapses following early feeding. There is strong evidence that enteral feeding is preferable to whole parenteral nutrition for patients with severe acute pancreatitis (Yeung et al., 2006). There have been reports of multiorgan failure, death, and local and systemic infection with total parenteral nutrition (Al-Omran et al., 2010; Chebli et al., 2005; Wu & Banks, 2013). The nasogastric and nasojejunal channels can be used to provide enteral nutrition. In cases of severe acute pancreatitis, nasogastric feeding is safe and has similar results to nasojejunal feeding (Chebli et al., 2005).

Initial management

Acute pancreatitis is initially managed mostly supportively, with fluid replacement, electrolyte optimization, adequate caloric support, and the prevention, identification, and treatment of local and systemic consequences. In acute pancreatitis, the local and systemic inflammatory response causes fluid loss by vomiting, decreased oral fluid intake,

third-space fluid loss, and increased sweat and respiration losses. With acute pancreatitis, Ringer's lactate crystalloid solution can be used to restore lost fluid (Chang et al., 2013).

Antibiotic treatment

Various secondary infective complications arise because of the translocation of gut bacteria due to altered motility of the gastrointestinal tract. Infected pancreatic necrosis can be effectively treated by using antibiotics that are especially effective against Gram-negative bacteria, such as imipenem, clindamycin, piperacillin, fluoroquinolones, and metronidazole. It has been found that usage of imipenem dramatically decreased the frequency of infected pancreatic necrosis, and carbapenems are associated with a considerable reduction in mortality as compared to other intravenous antibiotics (Mourad et al., 2017; Wu et al., 2011).

Surgical intervention

Gallstone pancreatitis requires specific treatments in serious conditions and may require various types of surgeries like cholecystectomy and cholecystostomy.

Various guidelines advise that the procedure of cholecystectomy should ideally be carried out at the index admission for patients with mild acute gallstone pancreatitis who are suitable and should be done within 2 weeks. It has been found that early laparoscopic cholecystectomy has shortened hospital stays in patients with acute gallstone pancreatitis as compared to patients who underwent delayed surgery. Patients who are not suitable for cholecystectomy, ERCP, and endoscopic sphincterotomy can be treated. All individuals with pancreatitis due to gallstones should undergo common bile duct imaging to detect choledocholithiasis. Transabdominal ultrasonography and/or MRCP are two noninvasive techniques used in preoperative imaging, whereas intraoperative cholangiography enables real-time imaging of the common bile duct. Choledocholithiasis can be managed in two ways that are as follows: (1) using a single-stage approach, which involves a laparoscopic or open cholecystectomy with intraoperative cholangiography and common bile duct exploration; or (2) using a two-stage approach, which involves preoperative ERCP with or without ES followed by laparoscopic or open cholecystectomy (Dasari et al., 2013; Lyu et al., 2019; Villatoro et al., 2010).

Conclusion

Similar to any other reason for acute pancreatitis, gallstone-related pancreatitis is a deadly disease as it damages the pancreas and ultimately becomes the main underlying factor for diabetes mellitus and exocrine pancreatic insufficiency. In addition, the inflammatory response that is induced by acinar cell damage very often leads to the generation of various types of immune responses in other organs, leading to multiorgan damage and failure. In the current scenario, consumption of fatty foods and alcohol are on the rise, and as a result, there is an increase in the incidence of gallstone formation, acute pancreatitis, and complications that arise due to it. Gallstone-related pancreatitis predominates among all gallstone-related diseases. The current treatment and disease management strategies have a good impact on the patients, but better management of the disease is only possible if we have a clear and better understanding of the disease at the cellular and molecular levels. In the current chapter, we have had a vivid discussion on the possible pathological pathway on how the disease happens in each step based on the available research evidence, and most of them are based on animal studies that mimic the situation similar to human beings. Based upon the evidence-based facts, we conceptualize and try to create similar situations that probably happen during gallstone pancreatitis, like the ligation of the hepatopancreatic duct to mimic hepatopancreatic blockage caused by gallstones. This is also worth stating that real physiological situations might be far different than our understanding assumed based on the animal studies, so continuous research is required to develop better disease models for getting a clear understanding of the disease, which is important for disease management and better drug development.

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